

ABDUL RAZAK

PHD · STAFF DATA SCIENTIST · AI AND MACHINE LEARNING RESEARCH

Salem, United States | [Portfolio](#) | [Google Scholar](#) | [DBLP](#) | [ORCID](#) | [Patents](#) | [LinkedIn](#)

RESEARCH PROFILE

Researcher and data scientist working at the intersection of **probabilistic reasoning**, **multi-objective decision making** and **generative artificial intelligence**. Doctoral and post-doctoral research with IBM Research Labs Dublin and United Technology Research Centre produced exact and approximate algorithms for decision problems with imprecise preferences and conflicting objectives, published at AAAI, IJCAI, UAI, CP and ICTAI. Now leads the generative-AI research programme for enterprise wireless networks at Extreme Networks, with nine granted US patents and a record of translating research into production systems.

RESEARCH INTERESTS Multi-objective decision making under uncertainty · preference elicitation and inference · probabilistic graphical models and influence diagrams · grounded and retrieval-augmented generative AI · constraint optimisation · machine learning for networked systems.

EDUCATION

NOV 2010 – MAY 2014 CORK, IRELAND	PhD in Computer Science (Artificial Intelligence) National University of Ireland, Cork — <i>Cork Constraint Computation Centre</i> Thesis: <i>Reasoning with Imprecise Trade-offs in Decision Making under Certainty and Uncertainty</i> — exact and approximate algorithms for multi-objective constraint optimisation and influence diagrams. Funded by the IRCSET / IBM Enterprise Partnership Scheme.
JUN 2003 – MAY 2005 HYDERABAD	MSc in Mathematics with Computer Science Osmania University — <i>awarded the University Merit Scholarship</i>
JUN 2000 – MAR 2003 HYDERABAD	BSc in Mathematics, Statistics and Computer Science Osmania University

APPOINTMENTS

2020 – PRESENT SALEM, US	Staff Data Scientist Extreme Networks — <i>previously Cork, Ireland</i> - Lead generative AI for enterprise wireless networking: language-model reasoning composed with statistical signal processing and a knowledge graph of devices, sites and topology. - Designed autonomous agent workflows that diagnose client behaviour, roaming failure and device anomalies across thousands of access points and switches, and propose remediation. - Built multi-modal inference pipelines over telemetry, device health and client experience, cutting fault investigation from hours to seconds; transferred preference inference into personalised recommendations.
2017 – 2020 CORK, IRELAND	Data Scientist Johnson Controls - Led machine learning across the intrusion-alarm, access-control and retail-analytics lines, delivering supervised, unsupervised and probabilistic models into commercial products. - Applied relational probabilistic programming (BLOG) and Bayesian inference (PyMC3) to alarm classification and root-cause analysis; eight granted US patents issued from this work.
2014 – 2017 CORK, IRELAND	Science Foundation Ireland Post-Doctoral Fellow Insight Centre for Data Analytics, NUI Cork Research on probabilistic graphical models, clustering and influence diagrams for industrial decisions within a United Technology Research Centre programme; extended the doctoral line to lexicographic and weighted-average preference models with IBM Research Labs Dublin.
2005 – 2010 HYDERABAD	Lecturer in Mathematics Osmania University Taught algebra, analysis, calculus, probability and statistics to undergraduate cohorts.

PEER-REVIEWED PUBLICATIONS COMPLETE LIST ON GOOGLE SCHOLAR AND DBLP

- [C6] **Multi-objective Influence Diagrams with Possibly Optimal Policies.** R. Marinescu, A. Razak, N. Wilson. *In Proc. 31st AAAI Conference on Artificial Intelligence, 2017.*
- [C5] **The Comparison of Multi-Objective Preference Inference Based on Lexicographic and Weighted Average Models.** Anne-Marie, A. Razak, N. Wilson. *In Proc. 27th IEEE Int. Conf. on Tools with Artificial Intelligence (ICTAI), 2015.*
- [C4] **Computing Possibly Optimal Solutions for Multi-Objective Constraint Optimisation with Trade-offs.** N. Wilson, A. Razak, R. Marinescu. *In Proc. 24th International Joint Conference on Artificial Intelligence (IJCAI), 2015.*
- [C3] **Multi-objective Constraint Optimisation with Trade-offs.** R. Marinescu, A. Razak, N. Wilson. *In Proc. 19th Int. Conf. on Principles and Practice of Constraint Programming (CP), 2013.*
- [C2] **Multi-objective Influence Diagrams with Trade-offs.** R. Marinescu, A. Razak, N. Wilson. *In Proc. 28th Conference on Uncertainty in Artificial Intelligence (UAI), 2012.*
- [W1] **Multi-Objective Constraint Optimisation with Lexicographic Preference Models.** Anne-Marie, A. Razak, N. Wilson. *Workshop on Algorithmic Issues for Inference in Graphical Models (AIGM), 2015.*

GRANTED US PATENTS NINE GRANTED, 2019-2021 · JUSTIA

DECISION INTELLIGENCE AND MAINTENANCE REASONING

- [P9] **Interactive method to reduce the amount of trade-off information required from decision-maker in multi-attribute decision-making under uncertainty.** A. Razak, L. A. Deleris, R. Marinescu, N. Wilson. *Granted 2019.*
- [P8] **Virtual Maintenance Manager.** A. Razak, G. Subramanian, C. J. Donovan. *Granted 2019.*
- [P7] **Root-cause analysis of event occurrences.** D. Horgan, A. Razak, G. Bernardi. *Granted 2021.*

ALARM MANAGEMENT AND OPTIMISATION

- [P6] **Automated Alarm Panel Classification using Pareto Optimization.** A. Razak, M. Stewart, G. Subramanian. *Granted 2020.*
- [P5] **Systems and methods for managing alarm data of multiple locations.** A. Razak, sole inventor. *Granted 2021.*
- [P4] **Building security system with alarm lists and alarm rule editing.** M. Costello, F. Fala, J. R. Holliday, N. B. Pavel, A. Razak. *Granted 2021.*

BUILDING SECURITY ANALYTICS

- [P3] **Building security analysis system with site-independent signature generation for predictive security analysis.** M. Dziduch, C. J. Donovan, A. Razak. *Granted 2020.*
- [P2] **Rules-based method of identifying misuse of emergency fire exits using data generated by a security alarm system.** M. Dziduch, C. J. Donovan, A. Razak. *Granted 2020.*
- [P1] **Systems and Methods for Managing a Security System.** M. Dziduch, C. J. Donovan, A. Razak. *Granted 2020.*

HONOURS, FELLOWSHIPS AND FUNDING

- IEEE Senior Member, elevated May 2026.
- Science Foundation Ireland Post-Doctoral Fellowship, Insight Centre for Data Analytics, NUI Cork, 2014–2017.
- IRCSET / IBM Enterprise Partnership Scheme scholarship, full doctoral funding, NUI Cork, 2010–2013.
- Osmania University Merit Scholarship (MSc, 2003–2005); ranked 40th of ~30,000 in the entrance examination.

ACADEMIC SERVICE AND INVITED TALKS

- Reviewer, *Computers and Electrical Engineering* (Elsevier) — 10+ manuscripts in AI and networking.
- Reviewer, IEEE International Conference on Communications (ICC) — AI submissions, 2025 technical programme.
- Invited talks: Sapienza University of Rome; NUI Galway; NUI Cork; IBM Research Labs Dublin.
- Conference presentations: oral and poster sessions at AAAI, IJCAI, UAI, CP and ICTAI.

TECHNICAL COMPETENCIES

PROGRAMMING	Python, C++ (STL, Boost, Eigen), SQL, PySpark, PyFlink
MODELLING	PyMC3, BLOG, Swift PPL, ILOG CPLEX, LPSOLVE, scikit-learn
METHODS	Bayesian inference, probabilistic graphical models, constraint programming, LLM reasoning
PLATFORMS	AWS SageMaker and Athena, Azure Databricks, real-time streaming and batch pipelines
LANGUAGES	English, Hindi, Telugu, Urdu · Irish citizen since 2021